
Online Examination Report

Date:	21.07.2014
Attempts:	1
Examinee:	Kanghuli, Joel
Date of birth:	9/9/1990
Subject:	020-AGK
Result:	44,99 %
Time used:	00:00:12 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	3651		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
2	2759		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
3	4495		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
4	3657		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
5	2769		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
6	3630		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
7	2754		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	1,00	1,00
8	2888		21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	0,00	1,00
9	2449	21.2.3	Aeroplane: Wings, tail and control surfaces	0,00	1,00
10	2396	21.2.3	Aeroplane: Wings, tail and control surfaces	1,00	1,00
11	1577	21.2.3	Aeroplane: Wings, tail and control surfaces	1,00	1,00
12	2417	21.2.3.3	Loads, stresses flutter	0,00	1,00
13	220	21.2.4	Fuselage, landing gear, doors, floor, etc	0,00	1,00
14	1271	21.4	LANDING GEAR, WHEELS, TYRES, BRAKES	0,00	1,00
15	836	21.6.3	Aeroplane: Pressurisation and air conditioning system	0,00	1,00
16	25	21.6.3	Aeroplane: Pressurisation and air conditioning system	1,00	1,00
17	1005	21.8.2.2	Design, operation, system components, etc	0,00	1,00
18	809	21.8.2.2	Design, operation, system components, etc	1,00	1,00
19	790	21.10.7	Ignition circuits	1,00	1,00
20	1164	21.11.2.4	Turbine	0,00	1,00
21	2824	22.	INSTRUMENTATION	1,00	1,00
22	4025	22.	INSTRUMENTATION	0,00	1,00
23	2822	22.	INSTRUMENTATION	0,00	1,00
24	4464	22.	INSTRUMENTATION	0,00	1,00
25	3109	22.	INSTRUMENTATION	1,00	1,00
26	4028	22.	INSTRUMENTATION	0,00	1,00
27	3097	22.	INSTRUMENTATION	0,00	1,00
28	3100	22.	INSTRUMENTATION	0,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	2797	22.	INSTRUMENTATION	1,00	1,00
30	3424	22.	INSTRUMENTATION	1,00	1,00
31	3405	22.	INSTRUMENTATION	0,00	1,00
32	3014	22.	INSTRUMENTATION	0,00	1,00
33	2806	22.	INSTRUMENTATION	0,00	1,00
34	3112	22.	INSTRUMENTATION	0,00	1,00
35	3102	22.	INSTRUMENTATION	1,00	1,00
36	1123	22.2	MEASUREMENT OF AIR DATA PARAMETERS	0,00	1,00
37	694	22.2.4	Altimeter	1,00	1,00
38	1690	22.2.6	Airspeed indicator (ASI)	1,00	1,00
39	1140	22.2.6	Airspeed indicator (ASI)	1,00	1,00
40	87	22.2.6	Airspeed indicator (ASI)	0,00	1,00
Total:45%				18,00	40,00

Attachment 2: List of result by topic

Licence

Date: 17.07.2014

Examinee: Kanghuli, Joel

Location: Uganda Civil Aviation Authority

Your Result: 18,00 from 40,00 Points (45,00 %)

Topics	Amount of Questions	Points achieved	Maximum Points	Percent
21. AIRFRAME AND SYSTEMS, ELECTR., POWERPLANT	20	10,00	20,00	50,00
21.2. AIRFRAME	5	2,00	5,00	40,00
21.4. LANDING GEAR, WHEELS, TYRES, BRAKES	1	0,00	1,00	0,00
21.6. PNEUMATICS - PRESSU AND AIRCO SYSTEMS	2	1,00	2,00	50,00
21.8. FUEL SYSTEM	2	1,00	2,00	50,00
21.10. PISTON ENGINES	1	1,00	1,00	100,00
21.11. TURBINE ENGINES	1	0,00	1,00	0,00
22. INSTRUMENTATION	20	8,00	20,00	40,00
22.2. MEASUREMENT OF AIR DATA PARAMETERS	5	3,00	5,00	60,00
Total	40	18,00	40,00	45,00

1

The purpose of a kill switch is to

- ☐ ground the battery eliminating current for the ignition system.
- ☒ ground the lead wire to the ignition coil shutting down the powerplant.
- ☐ shut off the fuel to the carburetor.

2

The first indication of carburetor ice in an aircraft with a fixed-pitch propeller is

☐ an increase in manifold pressure.

☒ a decrease in manifold pressure.

☐ a decrease in RPM.

3

What is the CFR Part 1 definition of critical engine?

- ☒ **The engine still running which would most adversely affect performance if it should fail.**
- ☐ **The outboard engine whose failure would most adversely affect safety.**
- ☐ **The engine whose failure would most adversely affect airplane performance or handling qualities.**

4

Adding more oil to the fuel than specified by the manufacturer of a 2-cycle engine will result in

- ☒ **increased carbon buildup and engine fouling.**
- ☐ **increased engine lubrication and optimal performance.**
- ☐ **increased engine performance.**

5

Running a fuel tank dry before switching tanks is not a good practice because

- ☐ any foreign matter in the tank will be pumped into the fuel system.
- ☒ the engine-driven fuel pump or electric fuel boost pump draw air into the fuel system and cause vapor lock.
- ☐ the engine-driven fuel pump is lubricated by fuel and operating on a dry tank may cause pump failure.

6

The purpose of the fuel tank vent system is to

- ☐ ensure a proper fuel to air ratio.
- ☒ allow air to enter the tank as fuel is consumed.
- ☐ remove dangerous vapors from the aircraft and prevent an explosion.

7

The presence of carburetor ice in an aircraft equipped with a fixed-pitch propeller can be verified by applying carburetor heat and noting

- ☐ a decrease in RPM and then a constant RPM indication.
- ☐ an increase in RPM and then a gradual decrease in RPM.
- ☒ a decrease in RPM and then a gradual increase in RPM.

8

A high-frequency vibration that suddenly occurs during flight could be an indication of a defective

☐ freewheeling unit.

☒ ignition system.

☐ transmission.

9

By changing the angle of attack of a wing, the pilot can control the airplane's

☒ lift and airspeed, but not drag.

☐ lift, airspeed, and drag.

☐ lift, gross weight, and drag.

Candidate: Joel Kanghuli
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Subject: 020-AGK
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10

Which of the following is considered a primary flight control?

☐ Dorsal fin.

☒ Elevator.

☐ Slats.

11

In a steep turn to the left, when using spoilers ...

- ☐ The right aileron will descend, the left one will ascend, the right spoiler will extend and the left one will retract.
- ☐ The right aileron will ascend, the left one will descend, the right spoiler will retract and the left one will extend.
- ☒ The right aileron will descend, the left one will ascend, the right spoiler will retract and the left one will extend.
- ☐ The right aileron will ascend, the left one will descend, the right spoiler will extend and the left one will retract.

12

What affects indicated stall speed?

- ☒ Load factor, angle of attack, and power.
- ☐ Angle of attack, weight, and air density.
- ☐ Weight, load factor, and power.

13

The fuselage of an aircraft consists, among others, of stringers whose purpose is to:

- ☒ withstand the shear stresses.
- ☐ integrate the strains due to pressurization to which the skin is subjected and convert them into a tensile stress.
- ☐ provide sound and thermal isolation.
- ☐ assist the skin in absorbing the longitudinal traction-compression stresses.

14

A tubeless tyre has : 1- a built-in-air tube. 2- no built-in-air tube. 3- a crossed side casing. 4- a radial side casing. The combination of correct statements is:

☒ 2, 3.

☐ 1, 4.

☐ 2, 4.

☐ 1,3.

15

If the cabin altitude rises (aircraft in level flight), the differential pressure:

- ☒ **increases**
- ☐ **may exceed the maximum permitted differential unless immediate preventative action is taken.**
- ☐ **decreases**
- ☐ **remains constant**

16

Assuming cabin pressure decreases, the cabin rate of climb indicator should indicate :

- ☐ a rate of descent of approximately 300 feet per minutes.
- ☐ a rate of descent dependent upon the cabin differential pressure.
- ☒ a rate of climb.
- ☐ zero.

17

During fuelling the automatic fuelling shut off valves will

- ☐ there is fire.
- ☒ fuelling system has reached a certain pressure.
- ☐ the fuel has reached a predetermined volume or mass.
- ☐ the surge vent tank is filled.

18

"Vapor lock" is the phenomenon by which:

- ☐ abrupt and abnormal enrichment of the fuel/air mixture following an inappropriate use of carburettor heat.
- ☐ burnt gas plugs forming and remaining in the exhaust manifold following an overheat and thereby disturbing the exhaust.
- ☐ water vapour plugs are formed in the intake fuel line following the condensation of water in fuel tanks which have not been drained for sometime.
- ☒ heat produces vapour plugs in the fuel line.

19

If the ground wire between the magnetos and the ignition switch becomes disconnected the most noticeable result will be that:

- ☐ the power developed by the engine will be strongly reduced
- ☐ the engine cannot be started with the ignition switch in the "ON" position
- ☐ a still operating engine will run down
- ☒ the engine cannot be shut down by turning the ignition switch to the "OFF" position

20

The control of free turbine engines on turboprops, is accomplished by: - a propeller control lever used to select: 1 - propeller RPM 2 - turbine temperature 3 - turbine RPM - a fuel control lever used to select: 4 - propeller RPM 5 - torque 6 - turbine temperature
The combination which regroups all of the correct statements is :

☐ 2 - 4 - 5

☐ 1 - 5 - 6

☐ 3 - 4 - 6

☒ 1 - 3 - 5

21

Prior to starting the engine, the manifold pressure gauge usually indicates approximately 29" Hg. This is because the

- ☐ throttle is closed, trapping high air pressure in the manifold.
- ☒ pressure within the manifold is the same as atmospheric pressure.
- ☐ pointer on the gauge is stuck at the full-power indication.

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22

In which publication can the VOR receiver ground checkpoint(s) for a particular airport be found?

☐ Aeronautical Information Manual.

☐ Airport/Facility Directory.

☒ En Route Low Altitude Chart.

23

A possible result of using the emergency alternate source of static pressure inside the cabin of an unpressurized airplane is the

- ☒ altimeter may indicate an altitude lower than the actual altitude being flown.
- ☐ airspeed indicator may indicate less than normal.
- ☐ altimeter may indicate an altitude higher than the actual altitude being flown.

24

(Refer to Figures 139 and 141.) Which displacement from the localizer centerline and glide slope at the 1,300-foot point from the runway is indicated?

- ☐ 21 feet above the glide slope and approximately 320 feet to the left of the runway centerline.
- ☒ 28 feet above the glide slope and approximately 250 feet to the left of the runway centerline.
- ☐ 21 feet below the glide slope and approximately 320 feet to the right of the runway centerline.

See Attachments:

- | | | |
|-----------------------|--------------------|-------------------|
| 1. instrument_139.jpg | instrument_139.jpg | (Attachment No.1) |
| 2. instrument_140.jpg | instrument_140.jpg | (Attachment No.2) |

25

Which is true concerning gyroplane operations?

- ☐ Rotor RPM will decrease during a vertical descent regardless of weight or density altitude.
- ☐ A gyroplane can take off from any area in which it can safely land.
- ☒ Minimum level flight speed is established when all available power is used to maintain constant altitude and airspeed.

26

When making an airborne VOR check, what is the maximum allowable tolerance between the two indicators of a dual VOR system (units independent of each other except the antenna)?

- ☒ **Plus or minus 4 degrees when set to identical radials of a VOR.**
- ☐ **6 degrees between the two indicated radials of a VOR.**
- ☐ **4 degrees between the two indicated bearings of a VOR.**

27

How can a pilot identify a lighted heliport at night?

- ☐ White and red beacon light with dual flash of the white.
- ☐ Green, yellow, and white beacon light.
- ☒ Green and white beacon light with dual flash of the white.

28

Identify the runway distance remaining markers.

- ☒ Red markers laterally placed across the runway at 3,000 feet from the end.
- ☐ Yellow marker laterally placed across the runway with signs on the side denoting distance to end.
- ☐ Signs with increments of 1,000 feet distance remaining.

29

Deviation error of the magnetic compass is caused by

- ☐ the difference in location of true north and magnetic north.
- ☒ certain metals and electrical systems within the aircraft.
- ☐ northerly turning error.

30

Which procedure is recommended to prevent or overcome spatial disorientation?

- ☐ Avoid steep turns and rough control movements.
- ☐ Reduce head and eye movements to the greatest extent possible.
- ☒ Rely entirely on the indications of the flight instruments.

31

How will the airspeed indicator react if the ram air input to the pitot head is blocked by ice, but the drain hole and static port are not?

- ☐ Indication will rise to the top of the scale.
- ☒ Indication will remain constant but will increase in a climb.
- ☐ Indication will drop to zero.

32

What is the significance of the blue radial line on the airspeed indicator of a light multiengine airplane and when is it to be used? It indicates the

- ☐ speed which will provide the maximum altitude gain in a given time when one engine is inoperative and should be used for climb and final approach during engine-out operations.
- ☒ minimum speed at which the airplane is controllable when the critical engine is suddenly made inoperative and should be used at all altitudes when an engine is inoperative.
- ☐ speed which will provide the greatest height for a given distance of forward travel when one engine is inoperative and should be used for all climbs during engine-out operations.

33

Which statement is true about magnetic deviation of a compass?

- ☐ Deviation varies for different headings of the same aircraft.
- ☒ Deviation is different in a given aircraft in different localities.
- ☐ Deviation is the same for all aircraft in the same locality.

34

What is the advantage of a three-bar VASI?

- ☐ A normal glide angle is afforded both high and low cockpit aircraft.
- ☒ Pilots have a choice of glide angles.
- ☐ The three-bar VASI is much more visible and can be used at a greater height.

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35

Which may lead to a power push-over in a gyroplane?

- ☐ Low speed.
- ☐ Decreasing power too quickly.
- ☒ Rotor force is removed.

36

A VSI may suffer from error due to:

- ☒ compressibility, time lag, blockage
- ☐ pitot blockage, static blockage, aircraft manoeuvre
- ☐ density, aircraft manoeuvre, position of the static vent
- ☐ time lag, instrument calibration, blockage

37

The high Altitude Radio Altimeter uses the following wavelengths:

☒ **decimetric.**

☐ **myriametric.**

☐ **hectometric.**

☐ **metric.**

38

VFE is the maximum speed:

- ☐ *with the flaps extended in the landing configuration.*
- ☐ *at which the flaps can be operated when flying in turbulences.*
- ☒ *with the flaps extended.*
- ☐ *with the flaps extended in the take-off configuration.*

39

An ASI consists of a gas tight casing which has:

- ☒ **pitot pressure fed into an open capsule and static pressure fed outside the capsule**
- ☐ **static pressure fed outside a sealed capsule**
- ☐ **static pressure fed outside a sealed capsule and pitot pressure fed into a second open capsule**
- ☐ **static pressure fed through a restrictive choke**

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40

Rectified Airspeed is:

- ☐ IAS corrected for density error.
- ☐ IAS corrected for instrument and pressure error.
- ☒ IAS corrected for density and compressibility errors.

Attachment No. 1

Questions 24

instrument_139.jpg

instrument_139.jpg

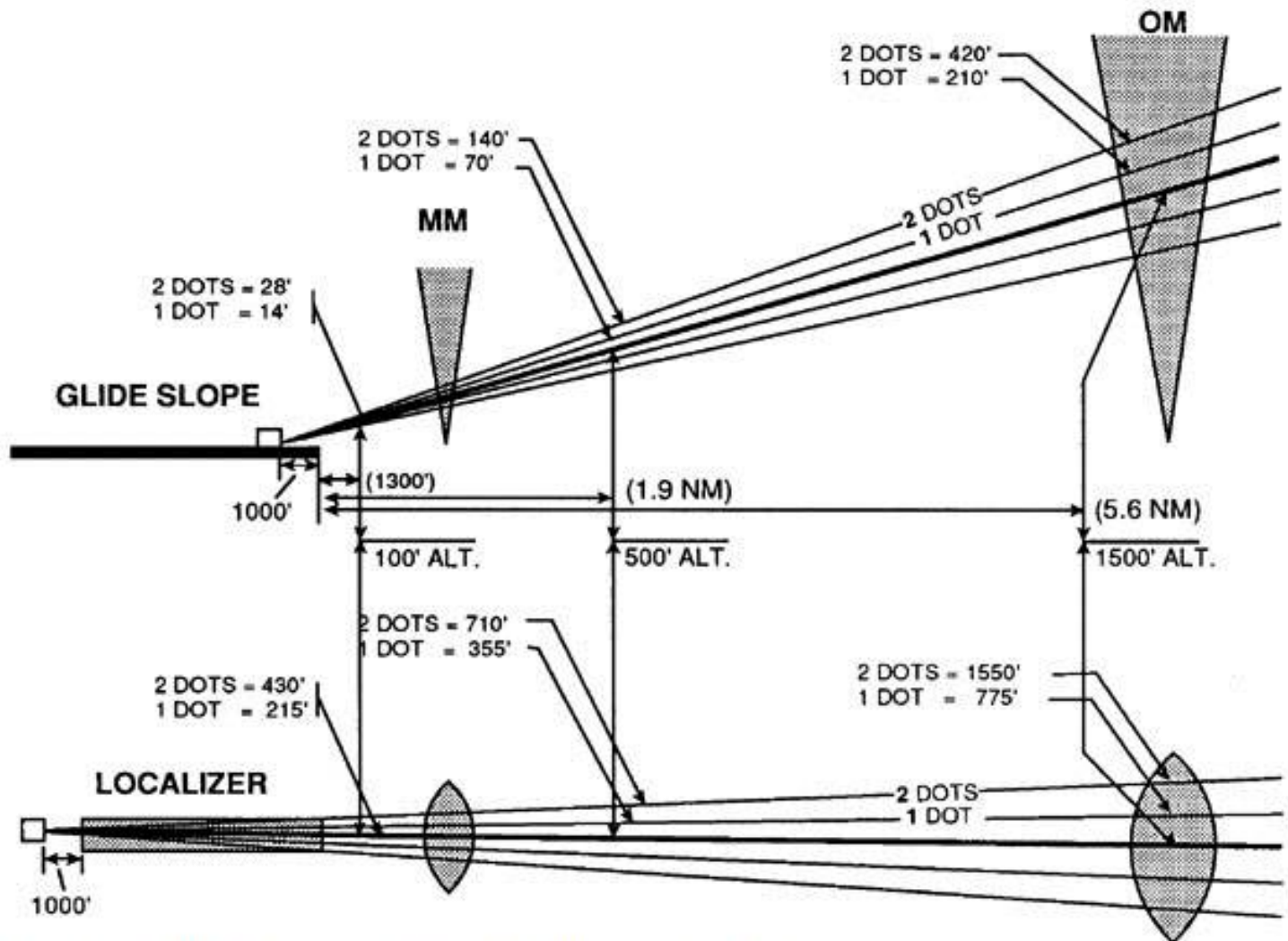


Figure 139. Glide Slope and Localizer Illustration © ASA

Attachment No. 2

Questions 24

instrument_140.jpg

instrument_140.jpg

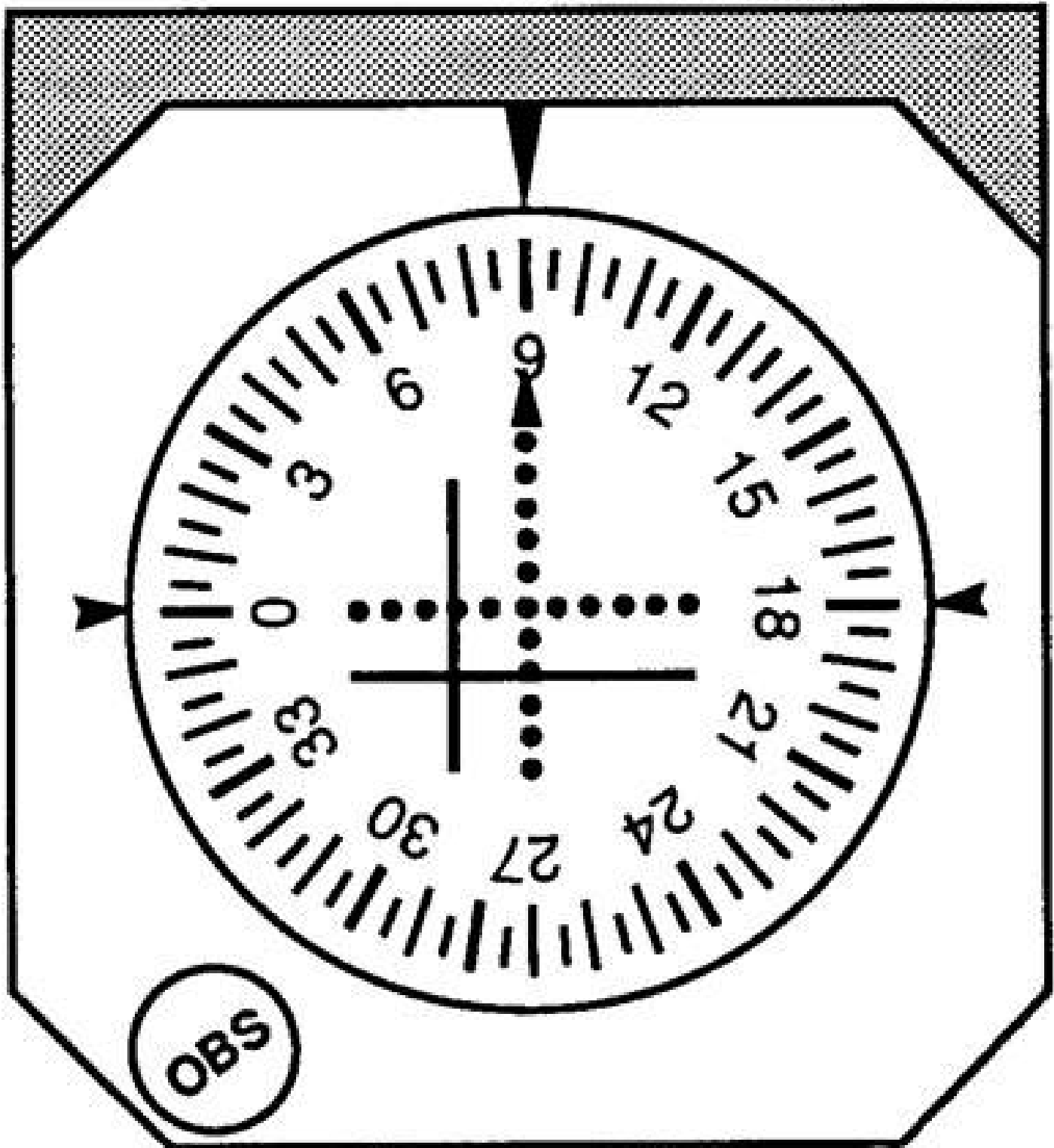


Figure 140. OBS, ILS, and GS Displacement © ASA